Ch. 15 Non-Textbook Questions

Principles of Microeconomics

July 31, 2026

1. Suppose that a market with 10 consumers and 5 producers is perfectly competitive. Each consumer's individual demand is given by $q_D = 40 - 2P$. Each firm has fixed costs of \$50 and marginal costs of $MC = \frac{q_S}{4}$. Marginal cost is greater than or equal to average variable cost for all $q_S \geq 0$.

   (a) What is the short-run market equilibrium, and how much does each firm produce?

   (b) What is each firm's profit at short-run equilibrium? Will firms enter or exit the market in the long run?

2. Newsprint (the paper used for newspapers) is produced in a perfectly competitive market. Each identical firm has a fixed cost of \$50, a variable cost of

$$VC = 40q + \frac{q^2}{2}$$

   and a marginal cost of

$$MC = 40 + q.$$

   (a) At what price will firms shut down in the short run?

   (b) There are 12 identical firms in this industry. Current market demand for newsprint is

$$Q_D = 360 - 2P.$$

   What is the short-run equilibrium price?

3. The raspberry growing industry in the United States is perfectly competitive, and each producer has a marginal cost curve given by

$$P = 20 + 2q,$$

   and an average total cost curve given by

$$P = 20 + q + \frac{144}{q}.$$

   The market demand curve is

$$Q_D = 2,488 - 2P.$$

   (a) What is the long-run equilibrium price in this industry?

   (b) How much does an individual firm produce at long-run equilibrium?

   (c) How many active producers are in the raspberry growing industry in a long-run competitive equilibrium?